1 Stiffness formulas

1.1 General

$k = \frac{F}{\Delta}$

Where:
- F is the force
- Δ is the deflection

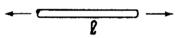
1.2 Torsional stiffness

$k = \frac{EI}{l}$

Where:
- E is the modulus of elasticity
- I is the moment of inertia of the cross-sectional area
- l is the total length

1.3 Axial force on a bar

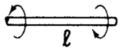
$k = \frac{EA}{l}$



Where:
- A is the cross-sectional area

1.4 Torque on a bar

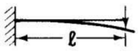
$k = \frac{GJ}{l}$



Where:
- G is the shear modulus or modulus of rigidity
- J is the moment of inertia

1.5 Force on the end of cantilever beam

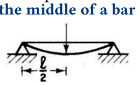
$k = \frac{3EI}{l^3}$



Note that k is at the position of the load.

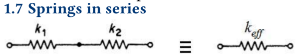
1.6 Force in the middle of a bar

$k = \frac{48EI}{l^3}$



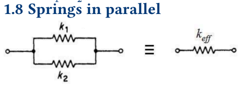
Note that k is at the position of the load.

1.7 Springs in series



$k_{\text{eff}} = \frac{1}{\frac{1}{k_1} + \frac{1}{k_2}}$,

1.8 Springs in parallel

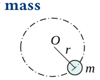


$k_{\text{eff}} = k_1 + k_2$

2 Moment of inertia formulas

2.1 Point mass

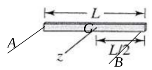
$J_o = mr^2$



2.2 Bar

$J_z = \frac{1}{12}mL^2$

$J_A = J_B = \frac{1}{3}mL^2$

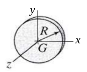


Note that z is the centre of the bar, and A and B are at the ends of the bar.

2.3 Disc

$J_z = \frac{1}{2}mR^2$

$J_x = J_y = \frac{1}{4}mR^2$

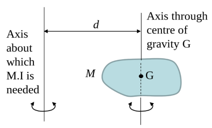


Note that z is the centre of the disc, and x and y are at the edge of the disc.

2.4 Parallel axis theorem

$J = J_G + Md^2$

Where:
- J is the moment of inertia needed
- J_G is the moment of inertia about its own centre of gravity
- M is the mass of the body
- d is the distance between the axes



3 Vibrations

3.1 Equations of motion

$m\ddot{x} + c\dot{x} + kx = 0$

$J_o\ddot{\theta} + c_\theta\dot{\theta} + k_\theta\theta = 0$

Where:
- m and J_o are the effective masses
- c and c_θ are the damping constants
- k and k_θ are the effective stiffness

3.2 Natural frequency

$\omega_n = \sqrt{\frac{\text{Effective stiffness}}{\text{Effective mass}}} = \sqrt{\frac{k}{m}} = \sqrt{\frac{k_\theta}{J_o}}$

$f_n = \frac{1}{2\pi}\sqrt{\frac{\text{Effective stiffness}}{\text{Effective mass}}} = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{k_\theta}{J_o}}$

3.3 Under-damped frequency

$\omega_d = \omega_n\sqrt{1-\zeta^2}$

3.4 Critical damping constant

$c_c = 2\sqrt{km} = 2m\omega_n = \frac{2k}{\omega_n}$

$(c_\theta)_c = 2\sqrt{k_\theta J_o} = 2J_o\omega_n = \frac{2k_\theta}{\omega_n}$

3.5 Damping ratio

$\zeta = \frac{c}{c_c} = \frac{c}{2\sqrt{km}} = \frac{c}{2m\omega_n} = \frac{c\omega_n}{2k}$

$\zeta = \frac{c}{(c_\theta)_c} = \frac{c}{2\sqrt{k_\theta J_o}} = \frac{c}{2J_o\omega_n} = \frac{c\omega_n}{2k_\theta}$

$\zeta = \sqrt{\frac{\delta^2}{4\pi^2+\delta^2}}$

3.6 Logarithmic decrement

$\delta = \ln\frac{x_1}{x_2} = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}}$

Where:
- x_1 is the amplitude of the first cycle
- x_2 is the amplitude of the second cycle

3.7 Static analysis

$\sum F = 0$

$\sum M = 0$

Static analysis is required when there is a spring with an unknown extension in static equilibrium, like it is counteracting the weight of the object.

3.8 Dynamic analysis

Newton's 2nd law:

$m\ddot{\theta} = \sum F \qquad J_o\ddot{\theta} = \sum M_o$

3.9 Undamped free vibrations

3.9.1 Equation of motion

$m\ddot{x} + kx = 0$

3.9.2 Displacement, velocity and acceleration

$x = X\sin(\omega_n t + \phi)$

$\dot{x} = \omega_n X\cos(\omega_n t + \phi)$

$\ddot{x} = -\omega_n^2 X\sin(\omega_n t + \phi)$

3.9.3 Amplitude

$X = \sqrt{x_0^2 + \left(\frac{\dot{x}_0}{\omega_n}\right)^2}$

3.9.4 Phase

$\phi = \arctan\left(\frac{x_0\omega_n}{\dot{x}_0}\right) + 180°$

3.10 Undamped forced vibrations

3.10.1 Equation of motion

$m\ddot{x} + kx = F_0\sin\omega t$

3.10.2 Displacement

$x = x_h + x_p$

$x = A\cos\omega_n t + B\sin\omega_n t + X\sin(\omega t + \phi)$

$x = X_h\sin(\omega_n t + \phi_h) + X_p\sin(\omega t + \phi_p)$

3.10.3 Steady-state solution

$x_p = X\sin(\omega t + \phi)$

3.10.4 Amplitude

$X = \frac{\frac{F_0}{k}}{1-\left(\frac{\omega}{\omega_n}\right)^2} = \frac{F_0}{k-m\omega^2}$

$\frac{X}{\left(\frac{F_0}{k}\right)} = \frac{1}{1-\left(\frac{\omega}{\omega_n}\right)^2}$

3.10.5 Phase

$\phi = 0$

3.11 Damped free vibrations

3.11.1 Equation of motion

$m\ddot{x} + c\dot{x} + kx = 0$

3.11.2 Under-damped

Damping ratio: $\zeta < 1$

$x = e^{-\zeta\omega_n t}X\sin(\omega_d t + \phi)$

$\dot{x} = -\zeta\omega_n e^{-\zeta\omega_n t}X\sin(\omega_d t + \phi) + e^{-\zeta\omega_n t}\omega_d X\cos(\omega_d t + \phi)$

$= -\zeta\omega_n x + e^{-\zeta\omega_n t}\omega_d X\cos(\omega_d t + \phi)$

$\ddot{x} = e^{-\zeta\omega_n t}X\sin(\omega_d t + \phi)(\zeta^2\omega_n^2 - \omega_d^2)$

$-2e^{-\zeta\omega_n t}\zeta\omega_d\omega_n X\cos(\omega_d t + \phi)$

$= x(\zeta^2\omega_n^2 - \omega_d^2) - 2e^{-\zeta\omega_n t}\zeta\omega_d\omega_n X\cos(\omega_d t + \phi)$

$\omega_d = \omega_n\sqrt{1-\zeta^2}$

3.11.3 Critically damped

Damping ratio: $\zeta = 1$

$x = (C_1 + C_2 t)e^{-\omega_n t}$

$\dot{x} = C_2 e^{-\omega_n t} - \omega_n(C_1 + C_2 t)e^{-\omega_n t}$

$= C_2 e^{-\omega_n t} - \omega_n x$

$\ddot{x} = \omega_n^2(C_1 + C_2 t)e^{-\omega_n t} - 2C_2\omega_n e^{-\omega_n t}$

$= x - 2C_2\omega_n e^{-\omega_n t}$

3.11.4 Over-damped

Damping ratio: $\zeta > 1$

$x = C_1 e^{r_1 t} + C_2 e^{r_2 t}$

$\dot{x} = C_1 r_1 e^{r_1 t} + C_2 r_2 e^{r_2 t}$

$\ddot{x} = C_1 r_1^2 e^{r_1 t} + C_2 r_2^2 e^{r_2 t}$

$r_1 = -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}$

$r_2 = -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}$

3.12 Damped forced vibrations

3.12.1 Equation of motion

$m\ddot{x} + c\dot{x} + kx = F_0\sin\omega t$

Unbalanced mass: $M\ddot{x} + c\dot{x} + kx = me\omega^2\sin\omega t$

Where:
- M is the effective mass, adding all the masses together
- m is the unbalanced mass
- e is the eccentricity

3.12.2 Displacement

$x = e^{-\zeta\omega_n}(A\cos\omega_d t + B\sin\omega_d t) + X\sin(\omega t + \phi)$

$\omega_d = \omega_n\sqrt{1-\zeta^2}$

3.12.3 Steady-state solution

$x_p = X\sin(\omega t + \phi)$

3.12.4 Amplitude

$X = \frac{F_0}{\sqrt{(k-m\omega^2)^2+(c\omega)^2}}$

At resonant frequency: $X_{\text{res}} = \frac{F_0}{\sqrt{(k-m\omega_n^2)^2+(c\omega_n)^2}}$

Unbalanced mass: $X = \frac{me\omega^2}{\sqrt{(k-M\omega^2)^2+(c\omega)^2}}$

3.12.5 Magnification factor or ratio

$\frac{X}{\left(\frac{F_0}{k}\right)} = \frac{1}{\sqrt{\left(1-\frac{\omega^2}{\omega_n^2}\right)^2+\left(\frac{2\zeta\omega}{\omega_n}\right)^2}}$

$\frac{X_{\text{res}}}{\left(\frac{F_0}{k}\right)} = \frac{1}{2\zeta}$ when $\omega = \omega_n$

Unbalanced mass: $\frac{MX}{me} = \frac{\left(\frac{\omega}{\omega_n}\right)^2}{\sqrt{\left(1-\frac{\omega^2}{\omega_n^2}\right)^2+\left(\frac{2\zeta\omega}{\omega_n}\right)^2}}$

$\frac{MX_{\text{res}}}{me} = \frac{1}{2\zeta}$ when $\omega = \omega_n$

3.12.6 Phase

$\phi = \arctan\left(\frac{c\omega}{k-m\omega^2}\right) = \arctan\left(\frac{\frac{2\zeta\omega}{\omega_n}}{1-\frac{\omega^2}{\omega_n^2}}\right)$

4 Maths

4.1 Small angle approximation

$\sin\theta \approx \theta$

$\cos\theta \approx 1$

$\tan\theta \approx \theta$

4.2 Derivatives

Chain rule:

$\frac{dz}{dx} = \frac{dz}{dy}\cdot\frac{dy}{dx}$

Product rule:

$\frac{d}{dx}(u\cdot v) = \frac{du}{dx}\cdot v + u\cdot\frac{dv}{dx}$

Quotient rule:

$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x)-f(x)g'(x)}{g(x)^2}$

Standard derivatives:

$\frac{d}{dx}(\sin x) = \cos x$

$\frac{d}{dx}(\cos x) = -\sin x$

$\frac{d}{dx}(\tan x) = -\sec^2 x$

$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$

$\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$

$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$

$\frac{d}{dx}(\sec x) = \sec x\tan x$

$\frac{d}{dx}(\csc x) = -\csc x\cot x$

$\frac{d}{dx}(\cot x) = -\csc^2 x$

4.3 Integrals

$\int\frac{1}{x} = \ln|x|$

$\int\sin x\,dx = -\cos x$

$\int\cos x\,dx = \sin x$

$\int\tan x\,dx = \ln|\sec x|$

$\int\frac{1}{x^2+a^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right)$

$\int\frac{1}{\sqrt{a^2-x^2}}\,dx = \arcsin\left(\frac{x}{a}\right)$

$\int\frac{1}{x^2-a^2}\,dx = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right|$

$\int\frac{1}{a^2-x^2}\,dx = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right|$

$\int\frac{1}{\sqrt{x^2-a^2}}\,dx = \ln\left|\sqrt{x^2-a^2}+x\right|$

$\int\sec x\,dx = -\ln|\sec x + \tan x|$

$\int\csc x\,dx = -\ln|\csc x + \cot x|$

$\int\cot x\,dx = \ln|\sin x|$

Integration by parts:

$\int u\,dv = uv - \int v\,du$

Preference for u:

1. L: log or ln.
2. I: Inverse trigonometric functions.
3. A: Algebraic functions.
4. T: Trigonometric functions.
5. E: Exponential functions.

4.4 Trigonometric identities

Quotient identities:

$\tan\theta = \frac{\sin\theta}{\cos\theta}$

$\cot\theta = \frac{\cos\theta}{\sin\theta}$

Reciprocal identities:

$\sin\theta = \frac{1}{\csc\theta}$

$\csc\theta = \frac{1}{\sin\theta}$

$\cos\theta = \frac{1}{\sec\theta}$

$\sec\theta = \frac{1}{\cos\theta}$

$\tan\theta = \frac{1}{\cot\theta}$

$\cot\theta = \frac{1}{\tan\theta}$

Pythagorean identities:

$\sin^2\theta + \cos^2\theta = 1$

$\sec^2\theta - \tan^2\theta = 1$

$\csc^2\theta - \cot^2\theta = 1$

Even/odd identities:

$\sin(-\theta) = -\sin\theta$

$\cos(-\theta) = \cos\theta$

$\tan(-\theta) = -\tan\theta$

$\cot(-\theta) = -\cot\theta$

$\csc(-\theta) = -\csc\theta$

$\sec(-\theta) = \sec\theta$

Co-function identities:

$\sin\left(\frac{\pi}{2}-\theta\right) = \cos\theta$

$\cos\left(\frac{\pi}{2}-\theta\right) = \sin\theta$

$\tan\left(\frac{\pi}{2}-\theta\right) = \cot\theta$

$\cot\left(\frac{\pi}{2}-\theta\right) = \tan\theta$

$\csc\left(\frac{\pi}{2}-\theta\right) = \sec\theta$

$\sec\left(\frac{\pi}{2}-\theta\right) = \csc\theta$

$\frac{\pi}{2}$ radians $= 90°$

Sum/difference identities:

$\sin(\theta\pm\phi) = \sin\theta\cos\phi \pm \cos\theta\sin\phi$

$\cos(\theta\pm\phi) = \cos\theta\cos\phi \mp \sin\theta\sin\phi$

$\tan(\theta\pm\phi) = \frac{\tan\theta\pm\tan\phi}{1\mp\tan\theta\tan\phi}$

Double angle identities:

$\sin(2\theta) = 2\sin\theta\cos\theta$

$\cos(2\theta) = \cos^2\theta - \sin^2\theta$

$\cos(2\theta) = 2\cos^2\theta - 1$

$\cos(2\theta) = 1 - 2\sin^2\theta$

$\tan(2\theta) = \frac{2\tan\theta}{1-\tan^2\theta}$

Half angle identities:

$\sin^2\theta = \frac{1-\cos(2\theta)}{2}$

$\cos^2\theta = \frac{1+\cos(2\theta)}{2}$

$\tan^2\theta = \frac{1-\cos(2\theta)}{1+\cos(2\theta)}$

Sum to product of 2 angles:

$\sin\theta + \sin\phi = 2\sin\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right)$

$\sin\theta - \sin\phi = 2\cos\left(\frac{\theta+\phi}{2}\right)\sin\left(\frac{\theta-\phi}{2}\right)$

$\cos\theta + \cos\phi = 2\cos\left(\frac{\theta+\phi}{2}\right)\cos\left(\frac{\theta-\phi}{2}\right)$

$\cos\theta - \cos\phi = -2\sin\left(\frac{\theta+\phi}{2}\right)\sin\left(\frac{\theta-\phi}{2}\right)$

Product to sum of 2 angles:

$\sin\theta\sin\phi = \frac{\cos(\theta-\phi)-\cos(\theta+\phi)}{2}$

$\cos\theta\cos\phi = \frac{\cos(\theta-\phi)+\cos(\theta+\phi)}{2}$

$\sin\theta\cos\phi = \frac{\sin(\theta+\phi)+\sin(\theta-\phi)}{2}$

$\cos\theta\sin\phi = \frac{\sin(\theta+\phi)-\sin(\theta-\phi)}{2}$

Law of sines:

$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Law of cosines:

$a^2 = b^2 + c^2 - 2bc\cos A$

Area of a triangle:

$A = \frac{1}{2}ab\sin C$